

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

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# Implementation of Naïve Bayes Algorithm in Python

# Import required libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score

# Load the Iris dataset
iris = load_iris()

# Separate features and target
X = iris.data
y = iris.target

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Naïve Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Calculate confusion matrix
cm = confusion_matrix(y_test, y_pred)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("Actual Values    :", y_test)
print("Predicted Values :", y_pred)

print("\nConfusion Matrix:")
print(cm)

print("\nAccuracy:")
print(accuracy * 100, "%")
```

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Actual Values    : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]
Predicted Values : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]
```

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Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Accuracy:
100.0 %